Костоломов Александр Евгеньевич
Модуль 1 Раздел 1

1. Подключился к серверу для выполнения практических работ, используя клиент для протокола ssh, командой
ssh -p 44556 eltex-pg1-v8@217.71.138.1

```
[zirael@MacBook-Air-RS ~ % ssh -p 44556 eltex-pg1-v8@217.71.138.1
[eltex-pg1-v8@217.71.138.1's password:
 Linux eltex-2026-spring 6.1.0-40-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.153-1 (2025-09-20) x86_64

 The programs included with the Debian GNU/Linux system are free software;
 the exact distribution terms for each program are described in the
 individual files in /usr/share/doc/*/copyright.

 Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
 permitted by applicable law.
 Last login: Wed Mar 11 16:57:03 2026 from 217.71.131.234
 eltex-pg1-v8@eltex-2026-spring:~$
```

2. Запустил новую сессию утилиты screen, с именем по умолчанию, командой screen

```
eltex-pg1-v8@eltex-2026-spring:~$
```

Посмотрел список подключенных пользователей командой w

```
eltex-pg1-v8@eltex-2026-spring:~$ w
 19:19:38 up 6 days,  7:57, 16 users,  load average: 0.01, 0.03, 0.00
USER     TTY      FROM             LOGIN@   IDLE   JCPU   PCPU WHAT
root     tty1     -                05Mar26  6days  0.02s   ?    -bash
tech     pts/1    37.193.179.176   05Mar26  6days  0.01s  0.13s sshd: tech [priv]
eltex-pg pts/3    37.195.178.14    16:52    1:38m  0.02s  0.02s -bash
eltex-pg pts/11   :pts/71:S.0      18:35   21.00s  0.03s  0.07s SCREEN -S eltex-pg2-v2_part2 -L -Logfile eltex-pg2-
eltex-pg pts/22   5.137.255.143    18:49    1:30   0.01s   ?    man less
eltex-pg pts/28   66.151.42.58     19:18    1.00s  0.01s   ?    screen
eltex-pg pts/41   :pts/28:S.0      19:18    1.00s  0.04s  0.04s w
eltex-pg pts/55   :pts/60:S.0      05Mar26  6days  1.04s  1.04s screen -r top
eltex-pg pts/60   :pts/38:S.0      05Mar26  6days  1.01s  1.01s screen -S top
eltex-pg pts/62   :pts/69:S.0      18:13   34:31   0.02s  0.02s script -f eltex-pg1-v10_part.log
eltex-pg pts/69   212.164.64.60    18:10   34:31   0.01s   ?    screen -d -r 1863243.eltex-pg1-v10_part2
eltex-pg pts/71   37.195.178.14    18:20   21.00s  0.02s  0.01s screen -S eltex-pg2-v2_part2 -L -Logfile eltex-pg2-
eltex-pg pts/72   :pts/21:S.0      05Mar26  6days  0.01s   ?    script -a eltex-pg1-v16_part.log
eltex-pg pts/84   :pts/55:S.0      05Mar26  6days  0.00s  0.01s SCREEN -S top -d -m
eltex-pg pts/121  :pts/82:S.0      05Mar26  6days  0.00s   ?    SCREEN -S top
eltex-pg pts/130  :pts/125:S.0     05Mar26  6days  0.00s   ?    SCREEN -S top -d -m
eltex-pg1-v8@eltex-2026-spring:~$
```

Отключился от сессии

```
[eltex-pg1-v8@eltex-2026-spring:~$ screen
[detached from 2017056.pts-28.eltex-2026-spring]
eltex-pg1-v8@eltex-2026-spring:~$
```

3. Запустил отсоединенную сессию утилиты screen, при этом запустил в этой сессии команду top, назвал сессию именем «top», командой по шаблону:

screen -S имя_сессии -d -m командная_строка

```
eltex-pg1-v8@eltex-2026-spring:~$ screen -S top -d -m top
```

```
top - 19:42:58 up 6 days,  8:20, 16 users,  load average: 0.04, 0.12, 0.07
Tasks: 439 total,   1 running, 430 sleeping,   8 stopped,   0 zombie
%Cpu(s):  0.4 us,  0.6 sy,  0.0 ni, 98.9 id,  0.0 wa,  0.0 hi,  0.1 si,  0.0 st
MiB Mem :   3914.7 total,   1835.9 free,    936.1 used,   1426.1 buff/cache
MiB Swap:    976.0 total,    976.0 free,      0.0 used.   2978.6 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
  40796 eltex-p+  20   0    6724   3500   2528 S   0.3   0.1   3:06.53 watch
  61187 eltex-p+  20   0   11640   5544   3400 S   0.3   0.1  12:07.42 top
  91120 eltex-p+  20   0    6184   2852   2432 S   0.3   0.1   1:29.55 watch
  92769 eltex-p+  20   0   11732   5664   3420 S   0.3   0.1  12:11.90 top
  96881 eltex-p+  20   0   11708   5616   3400 S   0.3   0.1  12:03.25 top
 104241 eltex-p+  20   0    6188   2944   2448 S   0.3   0.1   1:36.49 watch
 177893 eltex-p+  20   0    6188   2964   2528 S   0.3   0.1   1:32.42 watch
 264857 eltex-p+  20   0   11668   5348   3184 S   0.3   0.1  11:57.22 top
      1 root      20   0  168912  13480   9204 S   0.0   0.3   0:08.35 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.08 kthreadd
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_par_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 slub_flushwq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mm_percpu_wq
     11 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     13 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     14 root      20   0       0      0      0 S   0.0   0.0   0:26.05 ksoftirqd/0
     15 root      20   0       0      0      0 I   0.0   0.0   0:56.82 rcu_preempt
     16 root      rt   0       0      0      0 S   0.0   0.0   0:02.27 migration/0
     18 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     20 root      rt   0       0      0      0 S   0.0   0.0   0:02.57 migration/1
     21 root      20   0       0      0      0 S   0.0   0.0   0:24.63 ksoftirqd/1
     23 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
     24 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/2
     25 root      rt   0       0      0      0 S   0.0   0.0   0:02.56 migration/2
     26 root      20   0       0      0      0 S   0.0   0.0   0:25.89 ksoftirqd/2
     28 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/2:0H-events_highpri
     29 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/3
     30 root      rt   0       0      0      0 S   0.0   0.0   0:02.56 migration/3
     31 root      20   0       0      0      0 S   0.0   0.0   0:24.02 ksoftirqd/3
     33 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/3:0H-events_highpri
     38 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kdevtmpfs
     39 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 inet_frag_wq
     40 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kauditd
     41 root      20   0       0      0      0 S   0.0   0.0   0:00.25 khungtaskd
     42 root      20   0       0      0      0 S   0.0   0.0   0:00.00 oom_reaper
     43 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 writeback
     44 root      20   0       0      0      0 S   0.0   0.0   0:10.18 kcompactd0
     45 root      25   5       0      0      0 S   0.0   0.0   0:00.00 ksmd
     46 root      39  19       0      0      0 S   0.0   0.0   0:03.35 khugepaged
     47 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kintegrityd
     48 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kblockd
     49 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 blkcg_punt_bio
     50 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 tpm_dev_wq
     51 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 edac-poller
     52 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 devfreq_wq
     54 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:1H-kblockd
     55 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kswapd0
     62 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kthrotld
     64 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 acpi_thermal_pm
     66 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mld
     67 root       0 -20       0      0      0 I   0.0   0.0   0:00.46 kworker/2:1H-kblockd
     68 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 ipv6_addrconf
     73 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kstrp
     78 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 zswap-shrink
     79 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/u9:0
    124 root       0 -20       0      0      0 I   0.0   0.0   0:01.10 kworker/1:1H-kblockd
    134 root       0 -20       0      0      0 I   0.0   0.0   0:00.50 kworker/3:1H-kblockd
    153 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 ata_sff
    154 root      20   0       0      0      0 S   0.0   0.0   0:00.00 scsi_eh_0
    156 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 scsi_tmf_0
    157 root      20   0       0      0      0 S   0.0   0.0   0:00.00 scsi_eh_1
    158 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 scsi_tmf_1
    159 root      20   0       0      0      0 S   0.0   0.0   0:00.00 scsi_eh_2
    160 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 scsi_tmf_2
```

4. Получил список сессий, созданных утилитой screen, командой screen -ls

```
[eltex-pg1-v8@eltex-2026-spring:~$ screen -S top -d -m top
[eltex-pg1-v8@eltex-2026-spring:~$
[eltex-pg1-v8@eltex-2026-spring:~$ screen -ls
There are screens on:
        2047393.top     (03/11/2026 07:41:40 PM)        (Detached)
        2017056.pts-28.eltex-2026-spring        (03/11/2026 07:21:53 PM)        (Detached)
2 Sockets in /run/screen/S-eltex-pg1-v8.
[eltex-pg1-v8@eltex-2026-spring:~$ screen -r top
[detached from 2047393.top]
eltex-pg1-v8@eltex-2026-spring:~$
```

5. Подсоединился к сессии top командой screen -r top
6. Создал еще одно окно в сессии top комбинацией Ctrl-a c

```
eltex-pg1-v8@eltex-2026-spring:~$
```

7. В созданном окне выполнил команду:
watch /usr/bin/vmstat

```
Every 2.0s: /usr/bin/vmstat                                  eltex-2026-spring: Wed Mar 11 19:45:12 2026

procs -----------memory---------- ---swap-- -----io---- -system-- ------cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st
 1  0      0 1883068 156464 1303908    0    0     0     3   69  167  0  1 99  0  0
```

8. Получил список окон комбинацией Ctrl-a “

```
Num Name                                                                                    Flags

  0 top                                                                                         $
  1 bash                                                                                        $
```

9. Переименовал второе окно как «vmstat» комбинацией Ctrl-a A

```
Set window's title to: vmstat

Num Name                                                                                    Flags

  0 top                                                                                         $
  1 vmstat                                                                                      $
```

10. Перешел в окно, в котором работает top.

```
top - 19:47:17 up 6 days,  8:25, 17 users,  load average: 0.00, 0.05, 0.05
Tasks: 441 total,   1 running, 432 sleeping,   8 stopped,   0 zombie
%Cpu(s):  0.3 us,  0.5 sy,  0.0 ni, 99.2 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3914.7 total,   1834.3 free,    937.7 used,   1426.1 buff/cache
MiB Swap:    976.0 total,    976.0 free,      0.0 used.   2977.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
 244868 eltex-p+  20   0   11668   5428   3264 S   0.7   0.1  11:55.64 top
     21 root      20   0       0      0      0 S   0.3   0.0   0:24.64 ksoftirqd/1
  35427 eltex-p+  20   0   11748   5720   3476 S   0.3   0.1  12:16.29 top
  38091 eltex-p+  20   0    6724   3500   2528 S   0.3   0.1   3:08.22 watch
  42906 eltex-p+  20   0    6724   3416   2444 S   0.3   0.1   3:30.82 watch
  53191 eltex-p+  20   0   11720   5656   3416 S   0.3   0.1  12:28.15 top
  96881 eltex-p+  20   0   11708   5616   3400 S   0.3   0.1  12:03.62 top
1832239 eltex-p+  20   0   11640   5504   3360 S   0.3   0.1   0:13.58 top
1999071 root      20   0       0      0      0 I   0.3   0.0   0:00.87 kworker/0:0-events
2047394 eltex-p+  20   0   11792   5432   3280 R   0.3   0.1   0:00.71 top
      1 root      20   0  168912  13480   9204 S   0.0   0.3   0:08.35 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.08 kthreadd
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_par_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 slub_flushwq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mm_percpu_wq
     11 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     13 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     14 root      20   0       0      0      0 S   0.0   0.0   0:26.06 ksoftirqd/0
     15 root      20   0       0      0      0 I   0.0   0.0   0:56.84 rcu_preempt
     16 root      rt   0       0      0      0 S   0.0   0.0   0:02.27 migration/0
     18 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     20 root      rt   0       0      0      0 S   0.0   0.0   0:02.57 migration/1
     23 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
     24 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/2
     25 root      rt   0       0      0      0 S   0.0   0.0   0:02.57 migration/2
     26 root      20   0       0      0      0 S   0.0   0.0   0:25.90 ksoftirqd/2
     28 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/2:0H-events_highpri
     29 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/3
     30 root      rt   0       0      0      0 S   0.0   0.0   0:02.56 migration/3
     31 root      20   0       0      0      0 S   0.0   0.0   0:24.02 ksoftirqd/3
     33 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/3:0H-events_highpri
     38 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kdevtmpfs
     39 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 inet_frag_wq
     40 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kauditd
     41 root      20   0       0      0      0 S   0.0   0.0   0:00.25 khungtaskd
     42 root      20   0       0      0      0 S   0.0   0.0   0:00.00 oom_reaper
     43 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 writeback
     44 root      20   0       0      0      0 S   0.0   0.0   0:10.19 kcompactd0
     45 root      25   5       0      0      0 S   0.0   0.0   0:00.00 ksmd
     46 root      39  19       0      0      0 S   0.0   0.0   0:03.36 khugepaged
     47 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kintegrityd
     48 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kblockd
     49 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 blkcg_punt_bio
     50 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 tpm_dev_wq
     51 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 edac-poller
     52 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 devfreq_wq
     54 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:1H-kblockd
     55 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kswapd0
     62 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kthrotld
     64 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 acpi_thermal_pm
     66 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mld
     67 root       0 -20       0      0      0 I   0.0   0.0   0:00.46 kworker/2:1H-kblockd
```

11. Отсоединился от сессии top комбинацией Ctrl-a d

```
eltex-pg1-v8@eltex-2026-spring:~$ screen -r top
[detached from 2047393.top]
eltex-pg1-v8@eltex-2026-spring:~$
```

12. Запустил новую сессию командой screen и запустил в окне команду: vi report_part1.txt

```
~
~
~
~
~
~
~
~
"report_part1.txt" [New]                                                    0,0-1         All
```

13. Отсоединился от сессии (Ctrl-a d) и получил список сессий командой screen -ls

```
eltex-pg1-v8@eltex-2026-spring:~$ screen
[detached from 2057719.pts-28.eltex-2026-spring]
eltex-pg1-v8@eltex-2026-spring:~$ screen -ls
There are screens on:
        2057719.pts-28.eltex-2026-spring        (03/11/2026 07:48:13 PM)        (Detached)
        2047393.top     (03/11/2026 07:41:40 PM)        (Detached)
        2017056.pts-28.eltex-2026-spring        (03/11/2026 07:21:53 PM)        (Detached)
3 Sockets in /run/screen/S-eltex-pg1-v8.
eltex-pg1-v8@eltex-2026-spring:~$
```

14. Отключился от сервера, используя команду logout

```
eltex-pg1-v8@eltex-2026-spring:~$ logout
```

15. Подключился к серверу (как в п.1), и получил список сессий screen

```
eltex-pg1-v8@eltex-2026-spring:~$ logout
Connection to 217.71.138.1 closed.
zirael@MacBook-Air-RS ~ % ssh -p 44556 eltex-pg1-v8@217.71.138.1
eltex-pg1-v8@217.71.138.1's password:
Linux eltex-2026-spring 6.1.0-40-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.153-1 (2025-09-20) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Mar 11 19:21:06 2026 from 217.71.131.234
eltex-pg1-v8@eltex-2026-spring:~$ screen -ls
There are screens on:
        2057719.pts-28.eltex-2026-spring        (03/11/2026 07:48:13 PM)        (Detached)
        2047393.top     (03/11/2026 07:41:40 PM)        (Detached)
        2017056.pts-28.eltex-2026-spring        (03/11/2026 07:21:53 PM)        (Detached)
3 Sockets in /run/screen/S-eltex-pg1-v8.
eltex-pg1-v8@eltex-2026-spring:~$
```

16. Подключился к сессии с редактором и записал в файл "ФИО часть 1 модуля 1 успешно завершена дата и время"

```
Костоломов Александр Евгеньевич часть 1 модуля 1 успешно завершена 11.03.2026 19:53
~
~
~
```

```
Костоломов Александр Евгеньевич часть 1 модуля 1 успешно завершена 11.03.2026 19:53
~
~
~
```

```
"report_part1.txt" [New] 1L, 140B written
eltex-pg1-v8@eltex-2026-spring:~$
```

Закрыл все сессии screen.

```
eltex-pg1-v8@eltex-2026-spring:~$ screen -ls
No Sockets found in /run/screen/S-eltex-pg1-v8.

eltex-pg1-v8@eltex-2026-spring:~$
```